

Assignment_1_Introduction_to_R

George Gregory Fisher

2026-08

Welcome to the first assignment of Foundations of Data Science!

Instructions

- Write your code in the R code chunks provided. Make sure to comment your code.
- When you are finished with your assignment, upload this R Markdown file under the Assignments tab on our class canvas page.
- You can work with your classmates, but remember to write their name in the comments.
- If you are inspired by code you found online make sure you cite the website, comment your code, and make it your own. No copy and pasting.
- IMPORTANT: Explain what your code does in a sentence or two accompanying each answer. I want to know not just if your code works but if you understand WHY it works.

1) Below I have provided a list of random letters and numbers. Please conduct the following tasks on this list:

- a) What type of data is each of the “things” in the list? Write these types to a vector called “my_types”. Psssst here is a hint for how you might want to write this code: <https://www.geeksforgeeks.org/adding-elements-in-a-vector-in-r-programming-append-method/#>.
- b) Count how many of each data type appear in your vector.
- c) How many numbers in this list are greater than 5?
- d) Generate a factor of types from your “my_types” vector.
- e) Get the index (or indices) of the number “1” in the lst list.

```
lst <- list(4, 'j', 'Y', 5, 'f', 'K', 8, 'z', 'T', 'c', 0, 'B', 2, 'm', 'X', 9, 'd', 'V', 3, 'n', 'G', 1)

#lst <- list(4, 'j', "Y", 5, 1) ### Test list

# a) What type of data is each of the "things" in the list? Write these types to a vector called "my_types"

my_types <- c()
greater_than_five <- 0

for (i in seq_along(lst)){
  item_class <- class(lst[[i]])
  my_types <- append(my_types, values = item_class)
}

# length(my_types)

# b) Count how many of each data type appear in your vector.

counts <- table(my_types)
cat("Counts of number of types in vector 'my_types'", "\n")
```

Counts of number of types in vector 'my_types'

```
print(counts)
```

```
my_types
character    numeric
         40         21
```

c) How many numbers in this list are greater than 5?

```
for (i in seq_along(lst)){
  item_class <- class(lst[[i]])
  if (item_class == "numeric" & lst[[i]] > 5)
    greater_than_five <- greater_than_five + 1
}
```

```
cat("Number of 'numeric' type items in the list greater than five:", greater_than_five, "\n")
```

Number of 'numeric' type items in the list greater than five: 8

d) Generate a factor of types from your "my_types" vector.

```
my_types_factor <- as.factor(my_types)
```

```
cat("The number of levels in my 'my_types' vector are:", my_types_factor[1], "\n")
```

The number of levels in my 'my_types' vector are: 2

```
cat("The levels in my 'my_types' vector are:", levels(my_types_factor), "\n")
```

The levels in my 'my_types' vector are: character numeric

e) Get the index (or indices) of the number "1" in the lst list.

```
for (i in seq_along(lst)){
  if (lst[[i]] == 1)
    cat("The number 1 is at index:", i, "\n")
}
```

The number 1 is at index: 28

The number 1 is at index: 46

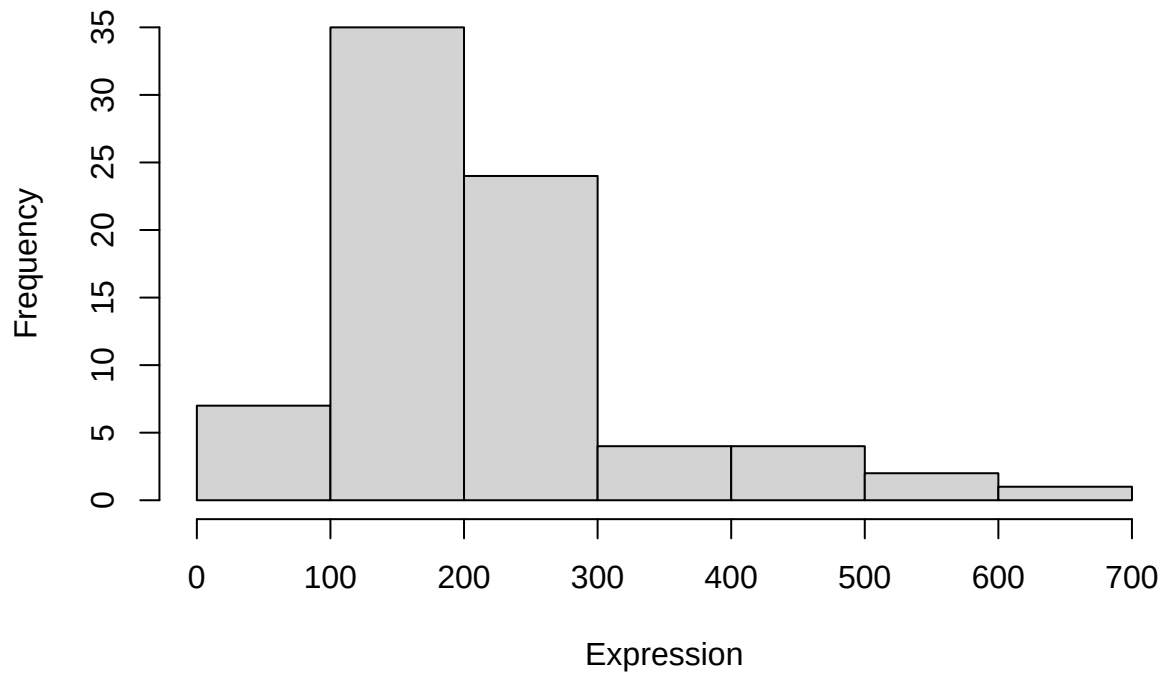
- 2) In the “data” folder of this repository there is a file called GSE165252_vst_PERFECT.txt. This is real RNA-seq data from a phase II clinical trial investigating the impact of combined impact PD-L1 inhibitor (atezolizumab) and neoadjuvant chemoradiotherapy on resectable esophageal adenocarcinoma patients. You can find a link to the data on GEO here: <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165252>. You can read about this study here: <https://aacrjournals.org/clincancerres/article/27/12/3351/671431/Neoadjuvant-Chemoradiotherapy-Combined-with>. Download the GSE165252_vst_PERFECT.txt file to your local machine.

- Read in this file. Make sure you take into account that the first row contains Ensembl IDs (gene names)!
- How many samples are in this data set?
- How many genes are in this data set?
- Make a histogram of the expression values of a gene of your choosing from this data set.
- Subset this data to contain only their first three samples and first 2000 genes. Write this subset to a file called “eso_sub.csv”.

```
## The number of genes in the 'rna_data': 62300
```

```
## The number of samples in the 'rna_data': 77
```

Histogram of ENSG00000000460.17



- 3) Write a function call “my_first_function”. The function should take the following vector: `my_vector <- c(1,2,3,4,5,6,7,8,9)`. Your function should sort the numbers in `my_vector` into two categories: less than or equal to 5 or greater than 5. Hint: You can take a look at for loops and if/else statements to complete this question!

```
my_vector <- c(1,2,3,4,5,6,7,8,9)

my_first_function <- function(vec,x) {
  for (val in vec) {
    if (val > x) {
      print(val)
    }
  }
}
```

Test the function

```
my_first_function(my_vector,5)
```

```
## [1] 6
## [1] 7
## [1] 8
## [1] 9
```